

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. in CSE Final Examination, 2022
CSE-2101: Data Structures and Algorithms

Total Marks: 70

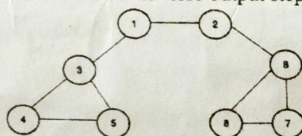
Time: 3 Hours

Answer any 5 (Five) of the following Questions:

1. a) (a) Write down the characteristics of Huffman coding. What is prefix rule? Explain with example. 4
- b) A file contains the following characters with the frequencies as shown. Find the Huffman Code for each character. prove that Huffman coding compresses the data set successfully. 5

Characters	Frequencies
a	10
e	15
i	12
o	3
u	4
s	13
t	1

- c) What is priority queue? Explain two types of priority Queue. Write down the application of Priority Queue. 5
2. a) Explain the Drawback of array implementation of queue and discuss about its solution 3
- b) Show the DFS traverse output steps of the following graph (starting node 1). 5



- c) Show all the steps to evaluate the following postfix expression using postfix expression evaluation algorithm: 1 2 3 * + 3 2 1 - + * 4
- d) List the application of tree. 2
3. a) Write down the basic operations of link list. When doubly linked list can be represented as circular linked list? 4
- b) Discuss a single link list. Write an algorithm to insert an item at the beginning of the list after a given node and at the end of the list. 5
- c) Mention the major drawback of doubly link list. Explain a step-by-step process for searching an item within an unsorted linked list. 5
4. a) What are the postfix and prefix forms of the expression given below $A+B*(C-D)/(P-R)$. 4
- b) Prove that the depth D of a complete binary tree with n nodes is given by $D = \lfloor \log_2 n + 1 \rfloor$, where $\lfloor . \rfloor$ is the floor function. 3
- c) All trees are Graphs, but all graphs are not trees. Explain this statement with example. 3
- d) What are the differences between DFS and BFS? 4
5. a) What do you mean by Tree? What are the steps to convert a general tree into a binary tree? 4
- b) Define Full Binary Tree. Generate the Tree for the following sequence:
Post order: F, A, E, K, C, D, H, G, B
In order: E, A, C, K, F, H, D, B, G 5
- c) What is Heap data structure? Build the Max-Heap and Min-Heap from the following list of numbers: 44, 30, 50, 22, 60, 55, 77, 55, 89, 19 5
6. a) Write down the Algorithm of insert operation in the max heap. 3
- b) If set A contains 13 elements, set B contains 8 elements and the intersection of these two sets contains 5 elements, then find the number of elements in A union B. 3
- c) Briefly explain asymptotic notation. 4
- d) Are the set $P = \{3, 8, 9\}$ and set $Q = \{9, 10, 11\}$, disjoint sets? If no justify your answer. 4
7. a) Define Sorting. Write down the pseudocode of the Selection sort algorithm. 4
- b) Compare Searching and Sorting. Describe the working procedure of Binary Search algorithm for the following data where the key value is 33.
10, 14, 19, 26, 27, 31, 33, 35, 42, 44 5
- c) What is insertion sort? How many passes are required for the elements to be sorted? 5